



Parallel Plate Capacitance and It's Distance Dependence

PHY-112

EXP-3

Group no:

Sec

ID

Name

Pre-Lab Questions

1. If a capacitor stores charge like a bucket holds water, does a bigger capacitor always fill up slower, just like a bigger bucket takes more time to fill? (Yes/No)
2. A 9V battery charges a capacitor. If we replace the battery with a 12V battery, will the capacitor charge faster, slower, or the same?
3. When you turn off a flashlight, the light goes out immediately. If a capacitor is like a tiny battery, does it also lose all its charge instantly when disconnected? (Yes/No)
4. If we use very thick wires instead of normal ones, will the capacitor charge noticeably faster?

Theory

A capacitor is used to store charge. A capacitor can be made with any two conductors kept insulated from each other. If the conductors are connected to a potential difference, V , as in for example the opposite terminals of a battery, then the two conductors are charged with equal but opposite amount of charge Q , which is then referred to as the "charge in the capacitor." The actual net charge on the capacitor is zero. The capacitance of the device is defined as the amount of charge Q stored in each conductor divided by the potential difference

$$C = Q/V$$

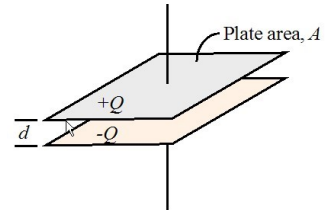
Rearranging gives:

$$V = Q/C \quad (1)$$

The simplest form of a capacitor consists of two parallel conducting plates, each with area A , separated by a distance d . The charge is uniformly distributed on the surface of the plates. The capacitance of the parallel-plate capacitor is given by:

$$C = \kappa \epsilon_0 A/d$$

Where κ is the dielectric constant of the insulating material between the plates ($\kappa = 1$ for a vacuum; other values are measured experimentally and can be found in tables), and ϵ_0 is the permittivity constant, of universal value $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$. The SI unit of capacitance is the Farad (F).



Equipment

INCLUDED:

1	Basic Electrometer	ES-9078
2	Basic Variable Capacitor	ES-9079
3	Electrostatics Voltage Source	ES-9077

NOT INCLUDED, BUT REQUIRED

1	550 Universal Interface	UI-5001
2	PASCO Capstone	
3	Paper	



Introduction

The purpose of this experiment is to investigate how the capacitance of a parallel-plate capacitor varies when the plate separation is changed and to qualitatively see the effect of introducing a dielectric material between the plates. A computer model of the system will be developed and the student will observe some of the power of computer modeling.

Setup

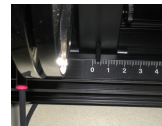
1. Move the Variable Capacitor plates so they are about 2 mm apart. Use the adjustment screws on the back of the moveable plate to make the plates parallel. Easiest way to do this is to look directly down from above the plates and adjust the horizontal adjust until the gap looks uniform, then look at the gap from the side and ever with the center of the plates and adjust the vertical screw. May need to repeat the process a few times.
2. Position the movable plate so the leading edge of the indicator foot (see Fig. 3) is at the 0.2 cm position. The gap between the two plates should be 0.2 mm all the way around. Check it with a ruler. If the gap varies repeat step 1. If the gap is not 0.2 mm, release the holding screw on the non-moving plate and move it until the gap is 0.2 mm and then tighten the screw back down.
3. Attach the twin lead (red & black) connector to the Signal Input jack on the Basic Electrometer. Route the wires as far away from where your hand and your body will be as possible. The charges in this experiment all small, so static discharge will foul things up. Also, people are conducting plates and have a significant amount of capacitance. Yo can foul things up just by being close. It is best to make the fixed plate ground by attaching the black wire's spade lug to it. Attach the red spade lug to the terminal on th moving plate. The wire must be free to move when the plate moves.
4. If you have a black banana/banana wire (not included) attach it as shown from the common (com) terminal on the Electrostatic Voltage Source to the ground terminal on the Electrometer. Alternately, use the provided banana/spade wire and connect the spade lead to the terminal on the fixed plate where the other ground lead is already attached. Attach the red banana/spade lead to the +30V terminal and leave the spade end free. Plug in the transformer and apply power to the Electrostatic Voltage Source. Shift the switch on the back to the On position. The green Power On light should glow.
5. Use the supplied adaptor cable to attach from the Signal Output on the Electrometer to the A Analog Input on the 850 Universal Interface. It is important that it be the A input!



Figure 2: Setup

Basic Variable Capacitor

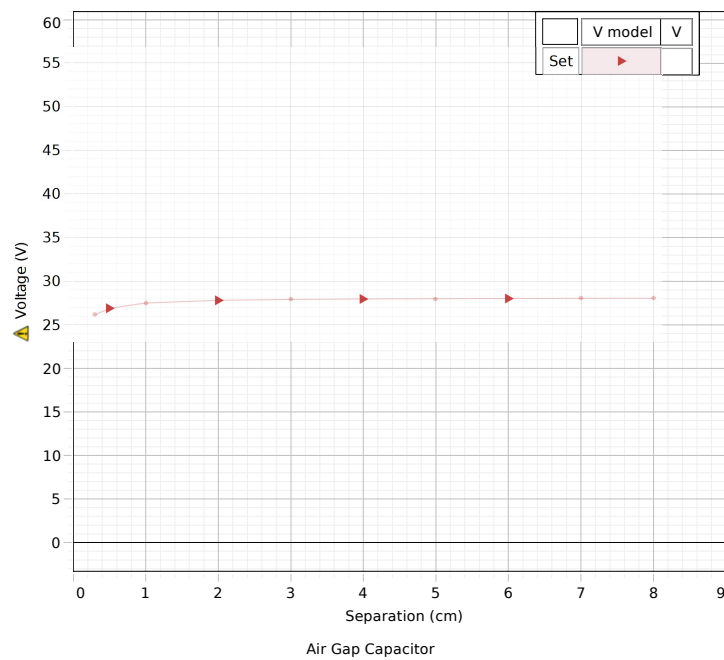
The PASCO experimental Variable Capacitor consists of two metal plates 17.7 cm (7 in) in diameter with a plate area $A = 2.46 \times 10^{-2} \text{ m}^2$.



[Figure 3: Position Indicator Foot]

Procedure: The Effect of the Plate Separation

1. Set the capacitor plates 0.3 cm apart by setting the movable plate so leading edge of its indicator foot is at the 0.3 cm mark.
2. Turn on the electrometer and set the range button to the 100 V scale.
3. Remove any charge from the capacitor by momentarily touching both plates at the same time with your hand.
4. Zero the electrometer by pressing the 'ZERO' button until the needle goes to zero.
5. Momentarily connect a cable from the +30V outlet in the voltage source to the stud on the back of the movable capacitor plate. This will charge the capacitor. Remove the charging cable.
6. Read the following steps. They need to be performed quickly since the charge will slowly escape from the electrometer, especially if the humidity is high. One person should run the computer while one moves the capacitor plate. Everyone else should stay back. Everyone should try to be in the same position for each reading. Anybody who is close is a significant part of the system and can make the readings change.
7. Open the Data Tab, but read the rest of this page first!
8. Slide the movable plate so it is at 8.0 cm (leading edge of the indicator foot). Once the plate is in position, the person moving the plate should move away 50 cm or so and try to be in the same position for each measurement.
9. Click the PREVIEW button at the lower left to begin collecting data. Colored numbers will appear in first row of the table. The person doing the computer should click the Keep Sample (red checkmark in the lower left) button. The number in the first row will turn black and the colored number will move to the second row. The person at the computer should read the next separation (7 cm) out loud and wait.
10. Move the plate to 8.0 cm and repeat the process until 0.3 cm.
11. Click the STOP button to end the data collection.
12. Examine the graph. If it looks like a smooth curve, you are done. If not, repeat the process until you get a nice looking run.

[illegible]

Analysis

$$V = Q/(\kappa\epsilon_0 A/d + C_{\text{sys}}) \quad (3)$$

Examination of Equation 3 (from Theory A) shows that if $C_{\text{sys}} = 0$, then V is directly proportional to d and the Voltage vs Separation graph on the Data page should be a straight line. This is clearly not the case. To verify Equation 3 for the case where C_{sys} is not zero, we need to know Q and C_{sys} . We determine these by fitting the math model (Equation 3) to the data.

First we note that: $\kappa\epsilon_0 A = (1.00)(8.85 \times 10^{-12} \text{ F/m})(2.46 \times 10^{-2} \text{ m}^2) = 2.18 \times 10^{-13} \text{ Fm} = 2.18 \times 10^{-11} \text{ F cm}$.

(So the parallel plate capacitance when $d = 1 \text{ cm}$ is $C_1 = 2.18 \times 10^{-11} \text{ F}$.) Note that this value is entered in line 2 of the Calculator.

When d is small (0.3 cm) the first term in the denominator dominates and

$$Q \sim V_{0.3}(\kappa\epsilon_0 A)/d = (30 \text{ V})(2.18 \times 10^{-11} \text{ F cm})/(0.3 \text{ cm}) = 2.2 \times 10^{-9} \text{ C}.$$

This value is entered as an initial guess for the value of Q in line 1 of the calculator. Q is constant since the plates are isolated, so when d becomes large, C_{sys} dominates in the denominator and we have:

$$C_{\text{sys}} \sim Q/V_8 \sim 2.2 \times 10^{-9} \text{ C}/80 \text{ V} = 2.7 \times 10^{-11} \text{ F}$$

Where V_8 is the voltage when $d = 8 \text{ cm}$. This is taken as the initial guess for C_{sys} ($= C_1$) on line 3 of the calculator. Note that C_{sys} is about equal to $C_{1.0}$ at 1.0 cm (see above). At 0.3 cm, $C_{0.3} = 7 \times 10^{-11} \text{ F}$ so $C_{0.3} \sim 3 C_{\text{sys}}$ and the approximation above is decent but not great. At 8 cm $C = 2.7 \times 10^{-11} \text{ F} = C_{\text{sys}}/10$, so the approximation is good, but not perfect.

1. Click open the Curve Fit page.
2. Use the Run Select button on the graph toolbar to select your best run.
3. Adjust the values for Q on line 1 of the Calculator and for C_{sys} on line 2 to make the model match the experimental curve as well as possible.
4. Answer the first four questions on the conclusions page.

Conclusions:

1. What happened to the voltage as the plates got closer together (d decreasing)?

2. What were your best fit values for the charge Q and Csys?

3. How well did your model fit the data? Try to explain any discrepancy. Hint: what approximations are made when deriving the parallel plate capacitance ($C = \epsilon_0 A/d$) from Gauss' Law?

4. Briefly discuss the value of computer modeling.